LP Gas Leak Detection Prototype for Rural Areas with AES-256 Encryption via LoRa and 3G Network Transmission.

line 1: Soto-Ortiz, S. I. [a], 2nd Salazar-Pérez, P. J. [b], 3rd Homero de Jesús Portillo Calva[c]
line 2a: [a]*Ingeniería en Tecnologías de la Información y Comunicaciones*
line 2b: [b]*Ingeniería en Tecnologías de la Información y Comunicaciones*
line 2c: [c]*Ingeniería en Tecnologías de la Información y Comunicaciones*
line 3: Tecnológico Nacional de México Campus Occidente del Estado de Hidalgo
line 4: Mixquiahuala de Juárez, Hgo., México
line 5: 1st ssoto@itsoeh.edu.mx, 2nd psalazar@itsoeh.edu.mx, 3rd 22011539@itsoeh.edu.mx

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***Abstract* - In Mexico, the widespread use of liquefied petroleum gas (LPG) in rural areas poses significant safety risks, accounting for 35.86% of domestic incidents reported by CENAPRED (2010–2020) [1]. This paper introduces "SafeGas," an Internet of Things (IoT) prototype designed for early LPG leak detection, integrating gas (MQ-6), pressure, and temperature sensors (MAX6675 with Type K thermocouple) alongside secure wireless communication via LoRa (Long Range) with AES-256 encryption and 3G (third-generation mobile networks).**

The prototype aims to monitor conditions in real time and trigger alerts through a dual-verification system (pressure + gas concentration), addressing key limitations of conventional leak detection methods. The development follows the IoT Development Lifecycle (IDL), structured into six phases: (1)Requirements: Defining objectives. (2)Design: System architecture. (3)Implementation: Hardware, firmware, and encryption. (4)Testing: LoRa range validation, false-positive rate assessment. (5)Deployment: Field installation in rural settings. (6)Maintenance: Firmware updates and performance optimization [24]

Keywords: Internet of Things (IoT), LoRa, AES-256, liquefied petroleum gas (LPG), encryption, 3G, MQTT.

I. Introduction

In Mexico, liquefied petroleum gas (LPG) is still the most common fuel in homes. It is useful and almost indispensable for cooking, but at the same time it represents a constant safety risk. Poor handling or unnoticed leaks can easily lead to fires, explosions, or even carbon monoxide poisoning [2]. Unfortunately, the country still lacks widespread and reliable systems to detect leaks, which explains the number of serious incidents that end in loss of lives and property.

Reports from CENAPRED covering 2010 to 2020 indicate that out of 1,425 domestic accidents, 914 happened in cities (64.14%) and 511 in rural areas (35.86%) [1]. This distribution is not random: higher population density, greater fuel consumption, and old infrastructure all play a role in the increased number of accidents in urban communities.

Several studies have investigated IoT solutions for gas leak detection and secure data transmission. González-González et al. [26] analyzed security in LPWAN networks like LoRa, emphasizing encryption to prevent tampering, while Herrera Rubio et al. [25] proposed a layered IoT model for Sigfox, highlighting reliability and data integrity challenges. Other efforts combined MQ-series gas sensors with Wi-Fi or GSM but often lacked encryption or had limited range, exposing data. These works reveal a gap in integrating multi-sensor verification with strong encryption and long-range communication, which the SafeGas prototype addresses.

SafeGas, tested in Mixquiahuala de Juárez, Hidalgo, uses AES-256-CBC to secure LoRa transmissions, keeping sensor data confidential and tamper-resistant. Its 256-bit key and block-chaining mechanism, processed in ~5.1 ms per block on the ESP32, also mitigate replay attacks. The system continuously monitors LPG concentration, pressure, and temperature, transmitting securely via LoRa with 3G as backup. Dual verification of gas and pressure reduces false alarms, while the low-cost, scalable design balances robust security with operational simplicity, providing an effective safety solution for remote communities. Figure 1 shows both nodes of the prototype and their joint operation.

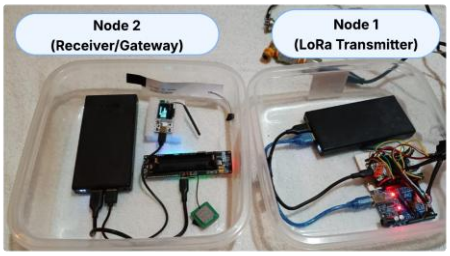


Fig. 1. Nodes of the physical prototype

II. State of the Art

In recent years, the use of the Internet of Things (IoT) for the detection of Liquefied Petroleum Gas (LPG) leaks has gained significant relevance. Kodali and Mandal [27] demonstrated the feasibility of using LoRaWAN for gas detection, achieving a range of 2 km in urban environments; however, their approach did not consider multi-sensor verification or security mechanisms for data transmission. Singh et al. [28] developed a wireless sensor network based on Wi-Fi and Node-RED for environmental monitoring using the MQ-6 sensor, achieving low latency but with a strong dependence on internet infrastructure, which limits its application in rural areas and lacks data protection measures.

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Regarding security in resource-constrained IoT devices, Perez et al. [29] implemented the AES-128 algorithm on ESP32 microcontrollers, confirming its feasibility with an encryption time of 3.8 ms per block; however, their work focused on agricultural applications and did not address the use of the more robust AES-256 standard or the context of home security. Meanwhile, Chen et al. [30] proposed the fusion of gas and pressure sensors for methane detection, significantly reducing the false alarm rate, although their system relied exclusively on LTE-M connectivity and presented high operational costs.

Unlike previous works, the SafeGas system combines multi-sensor verification (gas MQ-6, pressure, and temperature) with a hybrid communication architecture based on LoRa and 3G, designed to operate in rural environments without stable infrastructure. Furthermore, it implements end-to-end AES-256-CBC encryption, ensuring confidentiality and integrity with a processing time of 5.1 ms, allowing real-time operation. Altogether, SafeGas offers a comprehensive solution that unifies accurate detection, long-range communication, and advanced security, providing a significant contribution compared to previous approaches limited to a single sensor, Wi-Fi networks, or less robust encryption schemes.

III. Technologies And Tools Used

This section outlines the key technologies implemented in the project, covering both the IoT prototype's software and user interface components, along with all hardware elements:

A. *Software*

The system uses the **Arduino IDE** to program microcontrollers such as ATmega328P and ESP32 on Arduino Uno, Heltec LoRa 32 V3, and LilyGO TTGO T-SIM7600G-H R2 boards [3]. **Node-RED** manages IoT data flows, generates dashboards, and sends notifications via WhatsApp [4], [5]. Data is stored in **InfluxDB** [9] and visualized in **Grafana** [7], [8]. HiveMQ functions as the MQTT broker, receiving information from the LilyGO devices and forwarding it to Node-RED [6].

B. *Hardware*

The **Arduino Uno Rev3** reads sensors (**MQ-6, MAX6675, and a pressure transducer**) and transmits data via UART [10]. The **Heltec WiFi LoRa 32 V3** combines WiFi, Bluetooth, and LoRa, using the ESP32-S3FN8, to encrypt and transmit data over LoRa or decrypt and forward it via ESP-NOW [11], [17]. The **LilyGO TTGO T-SIM7600G-H R2** integrates 3G LTE, GPS, and WiFi, sending data to HiveMQ via MQTT [12], [18]. The **Raspberry Pi 4 Model B** functions as a local server running Node-RED, InfluxDB, and Grafana [13].

The sensors include the MQ-6 for LPG and methane (200–10,000 ppm) [14], the MAX6675 thermocouple (0–1024 °C) [15], and a pressure transducer (0–100/200 psi) [16]. Power is supplied via a 10,000 mAh power bank [22] to ensure continuous operation in rural areas.

C. *Communication protocols*

UART is used for Arduino–Heltec communication (<1 m, 9600–115200 bps) [3], **LoRa** for long-range connections between Heltec boards (up to 15 km, sub-1 GHz) [11], **ESP-NOW** for direct links between Heltec Node 2 and LilyGO (~200 m, latency <10 ms) [17], and **MQTT** to transmit data from LilyGO to HiveMQ via 3G with QoS and TLS support [18].

IV. Development

The prototype was deployed in rural homes, where the sensors were arranged in different points of the installation to improve the chances of detecting LP gas leaks. Figure 2 shows the physical layout of the system and how the elements were placed in areas considered critical for safety.

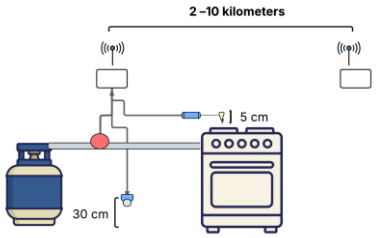


Fig. 2. Sensor distribution in a typical household environment

The system consists of two modules. The first, located near the gas tank, includes sensors, an Arduino, and a Heltec transmitter. It monitors the installation and sends encrypted data via LoRa, reaching 2–10 km depending on the environment. The second module, placed where mobile coverage is available, uses a Heltec receiver and a LilyGO 3G board to forward the data to the cloud.

Following NFPA 58 [23], three sensors are installed: a MAX6675 temperature sensor near the stove, a pressure sensor on the tank valve, and an MQ-6 gas sensor about 30 cm above the floor between the tank and stove, as LPG is heavier than air. This arrangement enables cross-checking: if the pressure sensor detects gas loss while the stove is off, a first warning is issued; if the MQ-6 also senses gas, a critical alarm triggers with sound and remote notifications. The dual verification reduces false alarms and improves detection reliability.

A. *IDL Methodology in the "SafeGas" Prototype*

The IoT Development Lifecycle (IDL) guided the prototype in six phases: requirements (dual detection and secure LoRa transmission), design (seven-layer architecture, ESP32, AES-256-CBC, MQTT), implementation (assembly and programming), testing (detection <4 s, LoRa range 2.8 km, 5.1 ms encryption), deployment (field trials in Mixquiahuala under NFPA 58), and maintenance (OTA updates and Raspberry Pi backups), resulting in a reliable, low-cost solution for rural households..

B. *Reference Model for IoT Architecture in LPG Leak Detection*

The proposed architecture follows a seven-layer IoT model inspired by works on Sigfox and LPWAN security [25], [26]. The lower layers (1–3) cover sensing and secure communication: an Arduino Uno acquires sensor data, which is transmitted via UART, LoRa with AES-256, ESP-NOW, and MQTT. Edge devices (Heltec, LilyGO) handle preprocessing—filtering, encrypting, and structuring—to reduce noise and improve efficiency.

The upper layers (4–7) ensure resilience and usability. A Raspberry Pi with HiveMQ provides local storage during internet outages, while Node-RED and JSON standardize interoperability. At the user level, Node-RED and Grafana dashboards enable real-time visualization, complemented by WhatsApp alerts for threshold exceedances. This design supports continuous monitoring, decision-making, and

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emergency response even in rural areas with unstable connectivity, surpassing conventional detectors.

C. Protocol Scenario

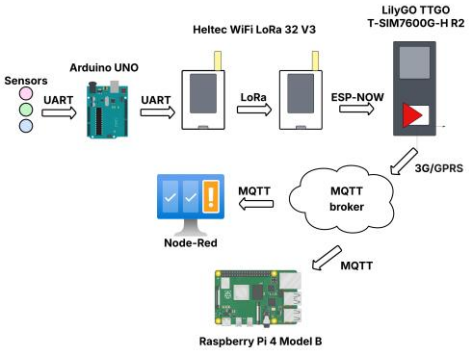


Fig. 3. Prototype architecture with protocols

Figure 3 shows the SafeGas LP monitoring system and its multi-node operation. An Arduino Uno collects data from three sensors. The measurements are formatted as a JSON object and sent via UART to the Heltec LoRa 32 V3 (Node 1), briefly displayed on its OLED screen, encrypted with AES-256-CBC using a fixed IV and shared key, and transmitted over LoRa to Node 2.

Node 2 decrypts the message, displays it locally, and forwards it via ESP-NOW to the LilyGO TTGO SIM7600 bridge. The LilyGO module publishes the JSON data to the HiveMQ GPL_Data topic over 3G/4G. Node-RED on a Raspberry Pi 4 B processes the stream, updating dashboards, and storing historical data in InfluxDB.

D. AES-256 Encryption in CBC Mode for LoRa Communications

AES-256 (Advanced Encryption Standard with a 256-bit key) in CBC mode (Cipher Block Chaining) is a widely trusted block cipher, often chosen for its balance between robustness and efficiency. At its core, the algorithm divides the input into 128-bit blocks (16 bytes). Each plaintext block is then combined—via an XOR operation—with the ciphertext of the block before it. The very first block, however, cannot rely on prior data; instead, it uses an Initialization Vector (IV) to introduce randomness and avoid predictable outputs [19]. To ensure that messages of any length fit the block structure, PKCS7 padding is applied. Table I outlines the key parameters used in this prototype.

TABLE I. KEY PARAMETERS OF AES-256 ENCRYPTION IN CBC MODE

Element	Value	Importance
Key	32 pre-shared bytes	Shared secret between transmitter/receiver
IV	16 fixed bytes	Should be unique for security (here fixed for simplicity)
Padding	PKCS7	Adjusts data to 16-byte multiples

Note: Parameters based on standard implementations [19], [21].

The encryption routine starts at Heltec Node 1, where sensor readings are formatted as JSON (e.g., {"gas":127,"pressure":52.38,"temp":24.50}) and padded with PKCS#7 to align to 16 bytes [20]. The padded message is then encrypted with AES-256-CBC and converted to a Base64 string, which alone is transmitted via LoRa—raw JSON never leaves the node. At Heltec Node 2, the process reverses: Base64 decoding, AES-256-CBC decryption using the same key and IV, and removal of padding restore the original JSON.

AES-256-CBC was chosen for its NIST standardization [31], widespread industrial and IoT adoption, cryptographic strength, and compatibility with embedded platforms like the ESP32 [32]. While newer algorithms like ChaCha20 [33] and Ascon [34] exist, they face challenges in low-power devices regarding native support and energy efficiency [35]. AES-256 provides an optimal balance of security, availability, and performance [36], ensuring data confidentiality and integrity over LoRa without affecting real-time operation. This approach enhances confidentiality, strengthens integrity via block chaining, and maintains efficiency on Heltec ESP32 boards. During prototyping, a fixed IV was used; production implementations must use a unique IV per session to maintain cryptographic robustness.

E. Physical Connections

Figure 4 shows a wiring diagram for Node 1 of the prototype.

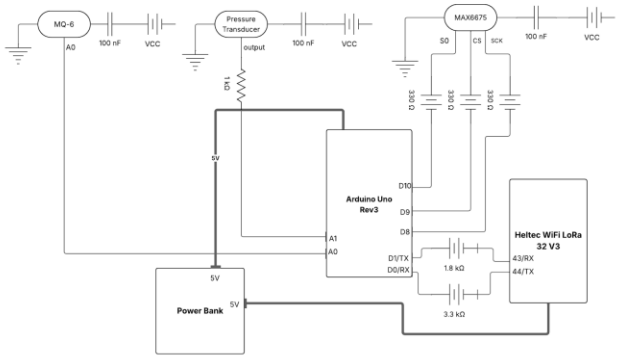


Fig. 4. Physical connections of the Arduino Uno and Heltec.

The prototype's power system operates with two nodes of distinct energy profiles. Node 1 (LoRa transmitter) consumes ~850 mA/h and is powered by a 10,000 mAh bank[22], supplying the Arduino Uno, MQ-6, pressure sensor, MAX6675, and Heltec LoRa 32 V3 for long autonomy in off-grid areas. Node 2 (receiver/gateway) consumes ~1.1 A/h due to its Heltec board and 3G/4G LilyGO module, enabling remote alerts. Sensor integration is handled by the Arduino: the MQ-6 and pressure sensor deliver analog signals to pins A0 and A1 (with filtering and decoupling), while the MAX6675 communicates via SPI (D8–D10 with protective resistors). Communication with the Heltec is via UART, with a voltage divider ensuring safe 5 V–3.3 V conversion. This dual-node design balances efficiency, redundancy, and reliable notifications, as validated in field trials.

F. Development Board Code

The Arduino Uno code implements a monitoring system with MQ-6 gas sensors, a pressure transducer, and a MAX6675 thermocouple. It calibrates the gas sensor in clean air to obtain a reference resistance (Ro) and, in the main loop, reads all

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sensors, converts signals to physical units (ppm, psi, °C), and transmits JSON-formatted data every second via UART.

The Heltec WiFi LoRa 32 V3 (UART Receiver and LoRa Transmitter) receives this JSON data, encrypts it with AES-256-CBC, transmits it via LoRa, and displays it on the OLED. Using LoRaWan_APP.h, HT_SSD1306Wire.h, and AESLib.h, communication occurs at 915 MHz with spreading factor SF7. Encryption employs a 32-byte key and 16-byte fixed IV; JSON data is byte-converted, PKCS#7-padded, and encrypted. The transmitted message combines IV and ciphertext, which the receiver separates to decrypt and recover the original JSON. Figure 5 illustrates this secure transmission routine.

Algorithm 1: *Secure Transmission Routine (AES-256-CBC + LoRa)*
Input: JSON_Message from UART
Output: Encrypted LoRa packet

```
Initialize UART, LoRa, OLED
Define AES256_KEY (32 bytes)
Define IV (16 bytes)
LOOP forever:
    if UART_Available():
        JSON_Message ← UART_Read()
        Padded_Data ← PKCS7_Pad(JSON_Message)
        Ciphertext ← AES256_CBC_Encrypt(Padded_Data, KEY, IV)
        Encoded_Message ← Base64_Encode(Ciphertext)
        LoRa_Send(Encoded_Message)
        OLED_Display("Encrypted data sent")
        if Transmission_Failed(): Retry(Encoded_Message)
        Manage_Power(OLED)
END LOOP
```

Fig. 5. AES-256 Encryption - Heltec Node 1

The main loop processes the UART data, encrypts it, and transmits it via LoRa while updating the OLED display. Transmission events confirm successful deliveries or manage retries, while auxiliary functions control the OLED power state to optimize energy consumption.

The Heltec WiFi LoRa 32 V3 ((LoRa Receiver and ESP-NOW Transmitter) code functions as a LoRa receiver that decrypts AES-256-CBC messages, displays values on the OLED, and forwards them via ESP-NOW to other ESP32 devices. It receives packets through LoRa, reads the RSSI, and sends the encrypted payload to the decryption module.

The security block uses AES-256-CBC with a 32-byte key and a 16-byte IV included in each message. The code extracts the IV, decrypts the ciphertext, and removes PKCS#7 padding to recover the original JSON, including fields like gas, pressure, and temperature.

After decryption, the program cleans and standardizes the data into JSON format, transmits it via ESP-NOW with a confirmation callback, and displays the processed values and transmission status on the OLED. Figure 6 illustrates receiving LoRa messages, decrypting AES-256-CBC, and sending data to the LilyGO gateway via ESP-NOW.

Algorithm 2: LoRa Receiver and ESP-NOW Transmitter
Input: Encrypted LoRa message
Output: JSON data sent over ESP-NOW

```
Initialize LoRa, OLED, ESP-NOW
Define AES256_KEY and IV (same as Node 1)
LOOP forever:
    if LoRa_Message_Available():
        Encoded_Message ← LoRa_Read()
        Ciphertext ← Base64_Decode(Encoded_Message)
        Decrypted_Data ← AES256_CBC_Decrypt(Ciphertext, KEY, IV)
        JSON_Message ← Remove_PKCS7_Padding(Decrypted_Data)
        ESPNOW_Send(JSON_Message)
        OLED_Display(JSON_Message)
        Confirm_Delivery()
END LOOP
```

Fig. 6. LoRa Receiver and ESP-NOW Transmitter - Heltec Node 2

The LilyGO TTGO (ESP-NOW Receiver and 3G Transmitter)code acts as an IoT gateway, receiving local data via ESP-NOW and forwarding it to the cloud through a 3G/LTE SIM7600 modem using MQTT. It integrates local reception, cellular communication, and connectivity management in a single device.

The SIM7600 setup includes modem initialization, SIM detection, and operator APN authentication (e.g., Telcel), adjustable for other networks. The MQTT client is configured with the HiveMQ broker and a custom topic for publishing. ESP-NOW reception allows the gateway to obtain JSON messages from nearby ESP32 devices, e.g., {"gas":120,"temp":25.5}, which are stored and prepared for remote transmission.

During initialization, the program connects to the cellular network, establishes GPRS, and sets up MQTT and ESP-NOW modules. In the main loop, it maintains connections, handles automatic reconnections, and publishes incoming data to the broker, ensuring continuous flow between the local network and the cloud. Figure 7 illustrates this process of receiving JSON via ESP-NOW and publishing it to HiveMQ over the 3G network..

Algorithm 3: ESP-NOW Receiver and 3G MQTT Publisher
Input: JSON_Message from ESP-NOW
Output: Published MQTT message to HiveMQ topic

```
Initialize ESP-NOW, SIM7600 modem, and MQTT client
Configure APN credentials and broker parameters (HiveMQ, topic “Gas_Data”)
LOOP forever:
    if ESPNOW_Message_Available():
        JSON_Message ← ESPNOW_Read()
        if not MQTT_Connected(): Reconnect_MQTT()
        MQTT_Publish(Topic="Gas_Data", Payload=JSON_Message)
        Log("Message sent to HiveMQ")
    Handle_Reconnection_And_KeepAlive()
END LOOP
```

Fig. 7. ESP-NOW Receiver and 3G MQTT Publisher - LilyGO

G. Node-red

The Node-RED flow receives sensor data via MQTT in JSON format, including temperature, pressure, and gas concentration. When gas levels exceed a threshold (1000–3000 ppm), an alert

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is triggered that sends a WhatsApp message with the current values and generates internal debugging traces.
The function node evaluates the thresholds and constructs the alert message, transforming raw data into useful notifications. This enables the user to quickly assess and respond to risks, which is crucial in rural areas with irregular infrastructure.
The Dashboard interface displays temperature, pressure, and gas level, providing immediate visual indicators that complement the automatic alerts and confirm the system's status.

H. Raspberry Pi 4 Model B (Local Server)

The Raspberry Pi 4 acts as a local server, centralizing the collection, storage, and visualization of sensor data. A **Mosquitto** broker, installed via Docker with the IoT Stack on port 1883, manages MQTT message reception and distribution, with sensors publishing to the Gas_Data topic.
Node-RED receives these messages, reformats the JSON data (e.g., renaming temp to temperature), and forwards it to **InfluxDB** on port 8086, storing it in the sensor_data database under the sensor_data measurement. Data integrity can be verified with queries like SELECT * FROM sensor_data.
Grafana then visualizes the data via dashboards configured with InfluxDB as the data source (http://<IP_Raspberry>:8086), generating interactive graphs for each sensor and offering customizable options for clear and immediate interpretation.

I. Physical Prototype

The circuit was assembled on a perforated board with soldered pins, allowing partial disassembly and quick sensor replacement—useful for field maintenance. The enclosure, 3D-printed from PLA using FDM, protects the circuit from impacts and external damage while providing openings for sensor and power wiring, preventing accidental disconnections during testing. Figure 8 shows the prototype inside the enclosure.



Fig. 8. The prototype inside the enclosure

V. Results

Tests were conducted in a controlled single-family household in Mixquiahuala de Juárez, Hidalgo, with stable environmental conditions (average 26 °C, moderate ventilation) and no deliberate changes in temperature, humidity, or airflow. Real LP gas was used via a regulated source. The system detected concentrations above 1,000 ppm with an average response time of 1.2 s, while high-risk levels (>10,000 ppm) triggered alerts and WhatsApp notifications in under 4 s.
AES-256-CBC encryption on the Heltec microcontrollers averaged 5.1 ms per block for encryption/decryption, and LoRa transmissions achieved an effective range of 2.8 km. Table II summarizes the key results.

TABLE II. Key results of the evaluations conducted on the "SafeGas" prototype in a simulated rural environment.

Evaluated Parameter	Obtained Value	Observation / Comparison with Related Works
Average detection time (≥1000 ppm)	1.2 seconds	The system activates the local sound alert. Comparable studies report 2–5 s; thus, SafeGas is more responsive.
Average AES-256 encryption/decryption	5.1 ms per 128-bit block	Evaluated on Heltec ESP32-S3 using AESLib. Ensures confidentiality and integrity with minimal delay.
Effective LoRa transmission range	2.8 km	Measured under rural line-of-sight conditions. Similar IoT systems typically report < 2 km in open areas, showing extended coverage.
MQTT sending latency (3G + HiveMQ)	2.3 seconds average	From LilyGO transmission to reception and dashboard visualization; acceptable for real-time monitoring.

Note: Data obtained from controlled tests. Comparable studies on LPWAN and IoT security and performance were reported by González-González et al. [26] and Herrera Rubio et al. [25].

The prototype demonstrates superior responsiveness and extended communication coverage while maintaining acceptable latency and strong cryptographic security. It is competitive among current IoT gas monitoring solutions for rural contexts.

VI. Conclusions

This work presented the development and implementation of an IoT prototype for LP gas leak detection in domestic environments, integrating gas, pressure, and temperature sensors with secure wireless communication via LoRa and 3G. Controlled and real-world tests demonstrated high detection performance: in 20 trials with LP gas concentrations above 1,000 ppm, the system triggered alerts in 18 cases (90% detection rate). The two remaining misses were linked to rapid gas dispersion caused by sudden air circulation changes, while only one false positive (5%) occurred, later reduced through a dual-validation scheme.

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AES-256-CBC encryption on embedded devices was validated, with an average processing time of 5.1 ms per 128-bit block, ensuring confidentiality and integrity without affecting operation. The prototype achieved an effective LoRa range of 2.8 km line-of-sight, confirming suitability for rural areas. Its low cost, scalability, and security-focused design make it a practical solution for accident prevention and remote monitoring. Future improvements include digital authentication (e.g., ECDSA), fault-tolerance, and support for distributed node networks to enhance robustness.

Limitations were noted: testing occurred in a single controlled household, limiting generalization; external factors like air currents, temperature changes, or electromagnetic interference could impact detection or LoRa stability; and using a fixed IV reduces cryptographic strength. These issues will be addressed in future iterations.

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